KARTIK TOMAR

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EDUCATION

Dr. A.P.J. Abdul Kalam Technical University — Lucknow
Bachelor of Technology in Computer Engineering — *2023 – Present*

East Delhi Public School — Ghaziabad
Senior Secondary (Class XII) - CBSE — *2022 – 2023*

East Delhi Public School — Ghaziabad
Secondary (Class X) - CBSE — *2020 – 2021*

PROJECTS

ResearchLab – AI Research Paper Assistant | *Python, Flask, RAG, Qdrant, Groq API, PyMuPDF*

- **RAG-Based Semantic Retrieval:** Built an AI research assistant using **384-dimensional embeddings**, **Qdrant vector search**, and **Groq LLM** for context-grounded Q&A with **page-level source citations**.
- **Cross-Paper Comparison:** Implemented paper-filtered retrieval to analyze **2–5 research papers** simultaneously, comparing methodologies, findings, limitations, and research gaps.
- **PDF Processing & Streaming:** Developed text, table, and figure extraction using **PyMuPDF and pdfplumber**; integrated **Server-Sent Events (SSE)** for real-time AI summaries and Q&A.

Deep Packet Inspection & Filtering Engine | *C++17, Multithreading, TCP/IP, PCAP, POSIX Threads*

- **High-Throughput Protocol Parsing:** Parsed Ethernet, IPv4, TCP, and UDP headers in native C++, achieving **∼2.5M packets/sec** parsing throughput with minimal memory allocation overhead.
- **Multi-Threaded Concurrency:** Engineered thread-safe worker pools using **std::condition_variable**, maintaining **>98% thread utilization** for efficient packet processing.
- **Stateful Flow Tracking:** Managed **100K+ simultaneous flows** through **5-tuple hashing**, ensuring consistent connection affinity across concurrent workers.

HireMe AI – Interactive AI-Powered Portfolio | *React, FastAPI, Groq API, Python, PyPDF, Tailwind CSS*

- **Conversational AI HR Agent:** Developed a macOS-inspired React portfolio with an embedded AI agent processing natural language queries with **sub-2-second** streaming response latency.
- **LLM-Powered JD Matcher:** Built a job description analysis engine delivering **95%+ fit-scoring accuracy** and skill-gap analysis across **10+ skill vectors**.
- **Automated Knowledge Base Sync:** Integrated **PyPDF** for dynamic resume parsing and context synchronization, enabling **100% automated updates** upon redeployment.

ACHIEVEMENTS & CERTIFICATIONS

- **OCI Agentic AI Foundations Associate:** Earned official credential from **Oracle** covering Agentic AI, LangChain, MCP, and LLM orchestration.
- **Policy Conclave'26 Finalist:** Selected as a Finalist at the **Indian Institute of Technology, Kanpur**.
- **Hacker's Premier League (Top 10):** Ranked in Top 10 at **Guru Gobind Singh Indraprastha University (Delhi)**.
- **National AI/ML Hackathon (Top 20):** Ranked in Top 20 in AVEVA Hackathon organized by **IIT Hyderabad** with Team Thunders **in Mar 2026**.
- **Competitive Programming:** Solved over **300+ coding problems** across platforms including **LeetCode**.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript.

Libraries & Frameworks: React.js, FastAPI, Flask, Node.js, Streamlit, NLTK, PyPDF, PyMuPDF, POSIX Threads (pthreads), HTML, CSS.

Developer Tools: Git, GitHub, VS Code, Qdrant, Docker, Render, Vercel, PCAP.

Coursework: Data Structures, Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks.

Soft Skills: Problem Solving, Communication, Decision Making.